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Logistics demand forecasting: a literature review

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Abstract

E-commerce has had significant growth in recent years, generating increased transportation demand and bringing challenges related to logistics resources and infrastructure planning. This paper develops a literature review on logistics demand forecasting, based on bibliometric analysis and systematic review of 48 selected papers. Results show that, when using a public sector point of view, there is a concentration of papers focused on long-term, high spatial aggregation models. The opposite happens when employing a private sector perspective, with more short-term and lower spatial aggregation. As a method, neural networks are gaining popularity. Additionally, mid-term logistics demand forecasting is little explored.

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Keywords: Logistics demand; logistic demand forecasting; urban logistics; city logistics.

1. Introduction

The COVID-19 pandemic has led to a significant growth in e-commerce around the world, generating an increasing demand for goods transportation and boosting the complexity of freight transportation systems and supply chains. These systems are composed of interconnected agents that perform the distribution of supplies through origin and destination flows that connect the supply side with demand side (Holguín-Veras & Jaller, 2014).

Transportation demand modelling is an important task from the point of view of public planners and logistics managers and the changes in the last years make it essential to study the logistic demand from a private point of view, so that companies can rely on predictive models to plan resources and optimize their logistic activities. Gonzalez-Feliu (2019) analysed the literature on transportation of goods and observed that in the 1970s this topic was addressed with the purpose of sizing the logistical structure of cities. In the following decade, route-based models met the interest of public authorities on traffic issues, and from the 90s on, personal mobility and freight transportation research emerged (Gonzalez-Feliu, 2019).

According to Ambrosini et al. (2008), urban goods movement models are scarce compared to passenger, vehicle traffic and interurban freight models. These authors showed that urban goods movement models published up to that time aimed at assisting the public planners' decision making. Data collection for these models were basically surveys

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on freight data, companies, residents, and socioeconomic data. Comi et al. (2012) identified four classes of urban freight models. The models were examined from the public planner point of view, for purposes such as traffic management and transportation planning.

Gonzalez-Feliu & Routhier (2012) examined the literature on urban freight focusing on approaches employed in demand estimation models. The authors pointed to the need to define model scope, as public planners and private companies have different objectives, leading to different choices regarding variables, granularity, and other parameters. Anand et al. (2015) define two categories of stakeholders for freight demand fulfilment in the domain of city logistics: infrastructure providers which are government authorities, and logistics service providers and companies that carry out transportation through a distribution network.

For private activities, objectives are usually related to cost, efficiency, and sustainability. Ballou (2006) cites examples of transportation decisions at the strategic (modal selection), tactical (seasonal equipment leasing), and operational (routing) levels, emphasizing the importance of planning due to the impact these decisions have on profitability. Last-mile deliveries represent approximately 41% of logistics costs and are at least twice as expensive as other supply chain costs, such as warehousing and sorting (Capgemini Research Institute, 2019).

On the other hand, the public sector is responsible for defining policies to reduce negative externalities produced by urban freight activity, in addition to planning the physical infrastructure for transportation, as discussed by Holguín-Veras et al. (2016). Among the goals of city logistics planners are improvements in economic output, reduction of road accidents linked to freight transport, reduction of the environmental impact of freight mobility, and equity (Nuzzolo & Comi, 2014). Initiatives with regional impact (as improving and creating new ring roads), metropolitan impact (such as freight clusters), areawide impact (such as exclusive lanes for trucks, vehicle weight and size restrictions) and local impact (as parking zones) were suggested by Holguín-Veras et al. (2020) for improving freight activity.

According to Holguín-Veras et al. (2011) and Holguín-Veras et al. (2013), freight generation (FG) models represent the amount of cargo to be transported (e.g: tons), and from this, the need for freight vehicle trips can be estimated through freight trip generation (FTG) models. The authors cite that these models should be treated separately, since FTG is not only dependent on FG, but also on shipment size: an increase in FG can lead to a less-than-proportional increase in FTG (Holguín-Veras et al., 2011).

Holguín-Veras & Jaller (2014) described the data requirements for different freight demand modelling techniques. The authors point out the different metrics to express freight demand, such as cargo value, the amount of cargo, number of trips, number of stops, and total deliveries. Holguín-Veras et al. (2013) found that the accuracy of FTG models is reduced due to incorrect model specification. The authors present a method to synthetically correct the model and provide more accurate estimates. In freight transportation modelling, different spatial scales can be addressed. Gonzalez-Feliu & Sánchez-Díaz (2019) define the macroscopic scale as urban areas or the overall demand of a city, the mesoscopic scale as neighborhoods or streets, and the microscopic scale as less aggregated areas or individual establishments.

Anand et al. (2015) categorized articles similarly to our approach, including stakeholders and descriptors. Other studies emphasized modelling, exploring the impact of aggregation levels on freight trip generation problem (GONZALEZ-FELIU, 2019). Our research explores logistics demand forecasting broadly, constructing a more extensive set of categories, from qualitative aspects of hierarchical planning to modelling methods and commonly used error metrics.

The main objective of this paper is to provide an overview of freight demand forecasting models by reviewing the current literature, and to discuss the profiles of models used for public and private activity. To this end, we developed a bibliometric analysis and classification of the articles based on a systematic review. Here, categories are empirically defined for the classification of articles. First, they were classified by target stakeholder (public planner or logistic manager), then by horizon (short-term or operational level: real-time, hourly, daily or weekly forecasts; mid-term or tactical: monthly forecasts; and long-term or strategical: annual or longer forecasts), spatial aggregation level (low: compatible with the mesoscopic scale; medium: compatible with the macroscopic scale; and high: scales larger than cities, such as metropolitan regions and provinces). Then, data sources, variables, methods, and evaluation metrics are identified for each article. Finally, we discuss the results and potential areas to be further explored in the future, as well as examine some relevant implications and opportunities.

2. Methodology

To conduct this literature review, the PRISMA (*Preferred Reporting Items for Systematic reviews and Meta-Analyses*) method is employed, with its four steps - identification, screening, eligibility, and inclusion (MOHER et al., 2009), see Fig. 1. In the first stage, we searched the titles, abstracts and keywords in the Scopus and Web of Science (WoS) databases, using two sets of words: i) “forecast" or "predict"; and ii) "logistic demand" or "logistic

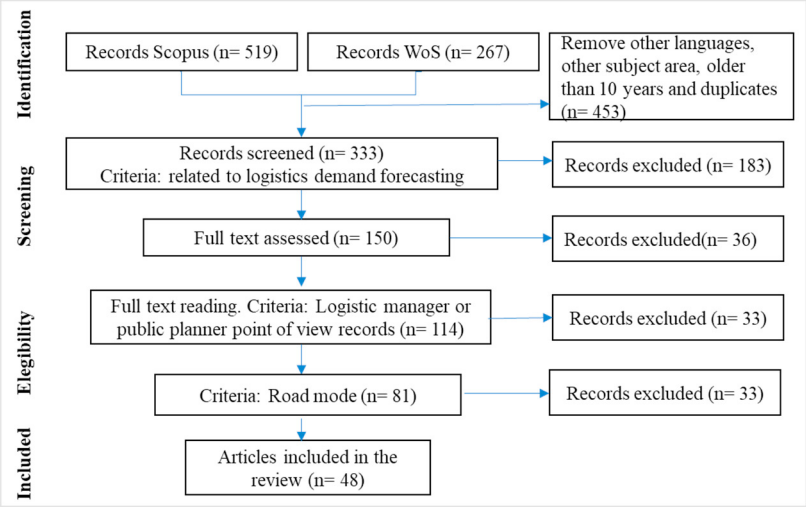

Fig. 1. PRISMA flow diagram

service" or "cargo volume" or "parcel deliver*" or "freight trip generation". We considered papers published in the last ten years (2013 to 2022), in English. For both databases, subject areas such as *Business*, *Computer Science*, *Decision Sciences*, *Economics*, *Engineering*, *Environmental Science*, *Mathematics*, *Transportation*, *Urban Studies*, and the like were researched. Duplicate papers were removed. Selected bibliography included journals and conference proceedings.

In the screening stage, two authors performed the screening of these works by reading the papers’ titles and abstracts, selecting those that addressed logistics demand forecasting, and that full access was available, resulting in 114 articles. The eligibility criterion adopted in the third stage was that the article should deal with logistics demand forecasting for public planners or logistics managers, and 81 articles were chosen by reading the full papers. Finally, 48 articles were included in the list for the literature review, based on an inclusion criterion that limits the list to works that deal with road transportation mode (more suited to urban logistics) and metric response variables. The metadata of the 48 articles came from the databases and entered Bibliometrix (Aria & Cuccurullo, 2017), and Vosviewer (van Eck & Waltman, 2010) tools to conduct the bibliometric analysis. Content analysis was performed to classify those articles according to the categories presented in the Introduction.

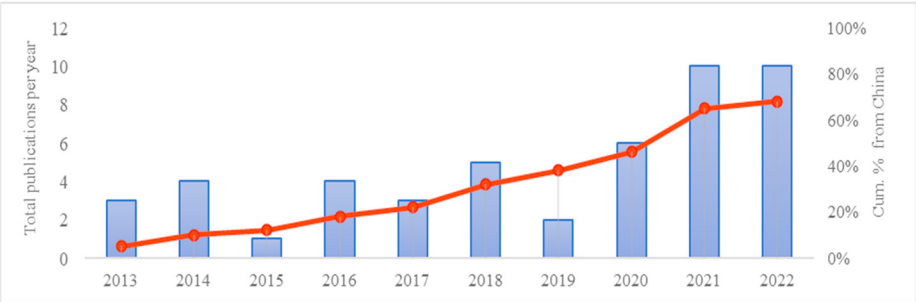

Fig. 2. Publications over time: total (bars); cumulative percentage from China (red line)

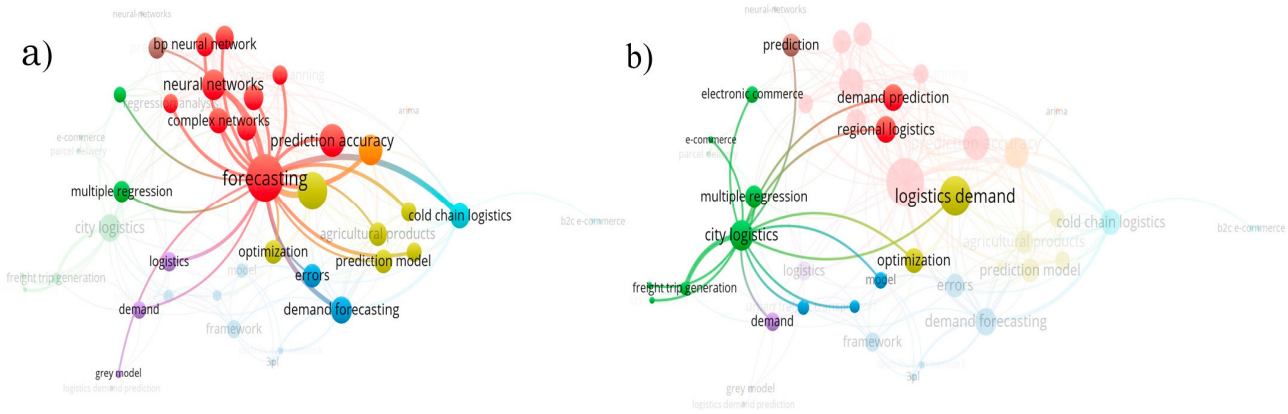

Fig. 3. Co-occurrence network. (a) forecasting links; (b) city logistics links

3. Results and discussion

After completing all steps of the PRISMA method, we proceeded to the bibliometric analysis of the 48 articles. Fig. 2 shows the evolution of publications and the accumulated percentage of Chinese publications over the last ten years. The sample presented an annual growth rate of 14.31%. A significant increase in the number of publications from 2020 onwards can be observed, showing that logistic demand forecasting gained increased attention.

When analyzing the scientific production by country, it was observed the predominance of China, which consistently leads the publications in this area, with 68% of the analysed articles. Brazil is the second country with the highest percentage of publications in the sample, with only 6%, demonstrating the great dispersion of publications by country. Regarding sources, *Sustainability* was the most relevant, with three publications. The remaining articles were published in 42 other journals and proceedings, a large dispersion.

The keywords co-occurrence network (Fig. 3) was created through the Vosviewer software. The circle size indicates the keywords relevance, among which we highlight *forecasting*, *city logistics*, *neural networks,* and others. Links between keywords show relationships between themes. *Forecasting* has a strong link to *cold chain logistics* and *logistics demand* and connects with methods that have appeared in the databases, such as *neural networks* and *multiple regression*. *City logistics* appears with a strong link to *freight trip generation* and connects to *regional logistics*, *demand prediction*, *logistics demand*, *optimization*, *multiple regression* and so on. There are no links between *forecast* and *city logistics*, indicating that the term *forecast* is not common in city logistics papers.

In the article classification stage, papers are evenly distributed between the logistic managers' point of view and the public planners', i.e., private, and public sectors' perspectives, respectively. Thus, Table 1 present the results of the classification of time horizon, spatial aggregation level, predicted variable, data source, methods and evaluation metrics concerning the logistic manager and public planner stakeholders.

Of the 24 articles related to the logistics manager's (LM) point of view (Table 1), 15 are classified as short-term: 13 are classified as low and 2 as medium spatial aggregation. Only one article is classified as medium term, and this one has medium aggregation. The other 8 articles are classified as long-term, being 7 with high spatial aggregation and the other one with medium aggregation. The variables weight, cubage and number of shipments were grouped into a single category called cargo volume which comprises half of all these papers. There are 5 papers focused on freight trip forecasting, and they all fall into the short term and low spatial aggregation category. Prediction models based on logistic indexes were found in 1 paper of medium and 2 of high spatial aggregation, with others 2 papers modeled delivery efficiency, being short-term and low spatial aggregation models. Interestingly, 2/3 of these papers applied primary data in the forecasting models (as actual historical data), all of them related to short and mid-term horizons. This is not a surprise, as companies have access to their own delivery data. About methods, neural networks (NN) were most frequent in short-term models, as well as time series (TS), which did not appear in other time horizons.

Table 1. Classification of the 48 articles identified in the literature review

Paper	Stakeholder	Horizon	Spatial aggregation	Predicted variable	Data source	Method(s)	Metric(s)
Yuan et al. (2018)	LM	Short-term	Low	Cargo volume	Primary	GM, NN	MAE, MSE
Ge et al. (2018)	LM	Short-term	Low	Cargo volume	Primary	FS	N/I
Kmiecik & Wolny (2022)	LM	Short-term	Low	Cargo volume	Primary	TS	MAE, MAPE
Sirikulvadhana et al. (2019)	LM	Short-term	Low	Cargo volume	Primary	TS	AIC, BIC
Kim et al. (2022)	LM	Short-term	Low	Cargo volume	Primary	NN	MAPE, MAE, RMSE
Liu & Pan (2019)	LM	Short-term	Low	Cargo volume	Primary	NN, TS	AIC, MRE
Shen & Lin (2020)	LM	Short-term	Low	Freight trip	Primary	NN	MAE, R^2, RMSE
Issaoui et al. (2022)	LM	Short-term	Low	Freight trip	Primary	NN	MAE, MAPE
Li et al. (2021)	LM	Short-term	Low	Freight trip	Primary	TS	N/I
Leung et al. (2020)	LM	Short-term	Low	Freight trip	Primary	FS, TS	R^2, RMSE
Nguyen Van et al. (2022)	LM	Short-term	Low	Freight trip	Primary	POI	N/I
Van Duin et al. (2016)	LM	Short-term	Low	Del. efficiency	Primary	REG	R^2
Praet & Martens (2020)	LM	Short-term	Low	Del. efficiency	Primary	DT	MAE
Zhou (2017)	LM	Short-term	Medium	Cargo volume	Primary	NN, TS	MSE
Kmiecik (2022).	LM	Short-term	Medium	Cargo volume	Primary	NN, TS	MAPE
Xue et al. (2018)	LM	Mid-term	Medium	Logistic index	Primary	GA, SVM	MAPE
Chen et al. (2020)	LM	Long-term	Medium	Cargo volume	Secondary	PCA, NN	N/I
Zhou (2013)	LM	Long-term	High	Cargo volume	Secondary	GM	MAE
Tian et al. (2021)	LM	Long-term	High	Cargo volume	Secondary	GM	RE
Xu & Lu (2021)	LM	Long-term	High	Cargo volume	Secondary	GM	RE
Lin & Lin (2021)	LM	Long-term	High	Cargo volume	Secondary	ES, GM, REG	MAE
Novotná & Gottwald (2021)	LM	Long-term	High	Cargo volume	Secondary	ES	N/I
Ma & Luo (2021)	LM	Long-term	High	Logistic index	Secondary	NN, REG	N/I
He & Yin (2021)	LM	Long-term	High	Logistic index	Secondary	GM, NN	MSE, RE
Alho & Silva (2014)	PP	Short-term	Low	Cargo volume	Secondary	REG	R^2, SE
Fadhlansyah & Nahry (2022)	PP	Short-term	Low	Cargo volume	Secondary	REG	R^2
Alho & de Abreu e Silva (2017)	PP	Short-term	Low	Freight trip	Secondary	REG	MAE, R^2, RMSE
Dhonde & Patel (2021)	PP	Short-term	Low	Freight trip	Secondary	REG	R^2, SE
Mepparambath et al. (2021)	PP	Short-term	Low	Freight trip	Secondary	REG	MAPE, MPE, R^2
Oliveira et al. (2022)	PP	Short-term	Medium	Freight trip	Secondary	REG	AIC, BIC, MAPE, RMSE
Dhonde & Patel (2020)	PP	Mid-term	Medium	Freight trip	Secondary	REG	R^2
Wang & Lu (2013)	PP	Mid-term	High	Cargo volume	Secondary	FS	MSE, RMSE
Sun et al. (2014)	PP	Long-term	Medium	Cargo volume	Secondary	EC, ES, GM, REG	N/I
Ya (2016)	PP	Long-term	Medium	Cargo volume	Secondary	NN, REG	MAPE, MSE
Wang & Wei (2018)	PP	Long-term	Medium	Cargo volume	Secondary	GM, REG	ADJUSTING R^2, R^2, SE
Huang et al. (2018)	PP	Long-term	Medium	Cargo volume	Secondary	NN	AE, FE, MSE
Yu, Xu & Yu (2020)	PP	Long-term	Medium	Cargo volume	Secondary	AC, SVM	MAPE, R^2, RE, RMSE
Liang & Geng (2013)	PP	Long-term	High	Cargo volume	Secondary	PSO, REG, SVM	MAE, MSE, MPE, RE, THEIL
He & Chang (2014)	PP	Long-term	High	Cargo volume	Secondary	GA, GM, REG	MSE, R^2
Hu et al. (2014)	PP	Long-term	High	Cargo volume	Secondary	NN	MAPE
Li & Lu (2015)	PP	Long-term	High	Cargo volume	Secondary	GM, REG	RE
Wang et al. (2016)	PP	Long-term	High	Cargo volume	Secondary	GM, NN	RE
Xia et al. (2017)	PP	Long-term	High	Cargo volume	Secondary	GM, NN	RE
Fu et al. (2022)	PP	Long-term	High	Cargo volume	Secondary	GM, NN	MAPE, RE
Zeng et al. (2022)	PP	Long-term	High	Cargo volume	Secondary	GM	RE
Jun et al. (2017)	PP	Long-term	High	Cargo volume	Secondary	KPCA, PSO, REG, SVM	MAE, MPE, RMSE, THEIL
Guo et al. (2021)	PP	Long-term	High	Cargo volume	Secondary	NN	MAPE, MSE, RE
Li & Wei (2022)	PP	Long-term	High	Cargo volume	Secondary	NN	MAE, RE, RMSE

AC= ant colony, Del = delivery, DT= decision tree, EC= elasticity coefficient, ES= exponential smoothing, FS= fuzzy system, GA= genetic algorithm, GM= grey model, KPCA= kernel principal component analysis, LM = logistic manager, NN= neural network, PCA= principal component analysis, PP = public planner, POI= Poisson process, PSO= particle swarm optimization, REG= regression, SVM= support vector machine, TE= trend extrapolation, TS= time series.

As for the articles from the public planner (PP) point of view (Table 1), 16 were categorized as long-term, of which 11 were classified as high spatial aggregation A total of 6 articles were classified as short-term and only 2 as mid-term. High spatial aggregation level is predominant, with 12 papers fitting in this category, another 7 as medium level and 5 as low level. The predicted variables identified were cargo volume (80% of the papers), and the rest considered the variable freight trips. It can be observed that all those classified as PP stakeholder have used secondary data to model the problem. These papers take an indirect approach to study factors that influence demand, such as economic indicators like gross domestic product (see He & Chang, 2014; Sun et al., 2014), industry output (Jun et al., 2017; Ya, 2016), and number of employees and establishment area (Oliveira et al., 2022). Regression models (REG) were used in 14 articles from short to long-term, while NN methods and Grey Prediction Models (GM) were applied in 8 articles each, all articles being related to long-term horizons.

In general, papers covered classical methods such as exponential smoothing (ES), REG, TS, and machine learning. Methods classified as NN include long short-term memory (LSTM), multilayer perceptron (MLP), backpropagation, convolutional networks, and others. Some papers employed more than one method, such as Yuan et al. (2018) and He & Yin (2021), who combined GM and NN to work with short-term, low spatial aggregation, and cargo volume prediction; and long-term, aggregated form and logistic index prediction, respectively. Only 33% of the papers have used primary data to generate estimates, all of them from the logistic manager's perspective (see Shen & Lin, 2020 and Li et al., 2021).

Classic model evaluation and comparison metrics have been widely used, such as relative error (RE), mean absolute error (MAE), mean percentage error (MAPE), mean absolute percentage error (MAPE), mean squared error (MSE), root mean squared error (RMSE), Akaike information criterion (AIC), Bayesian information criterion (BIC) and R^2. THEIL Index was used in only two papers. Tables 1 and 2 show in detail the classification of each article, and some differences between the publications profile from the logistic manager and public planner stakeholders perspectives become clearer.

Fig. 4 shows the classification of all 48 articles into orthogonal sectors representing different combinations of time horizon and spatial aggregation: when employing the public planner point of view, authors concentrate on long-term, higher spatial aggregation, while the opposite happens from the manager point of view.

Additionally, the concentration of articles in the Northeast and Southwest corners may not be a coincidence: forecasting is not an activity per se, i.e., it furnishes parameters and data for decision-making. Thus, the NE corner deals with strategic issues, as infrastructure and policy planning (public planner) and network design (private manager), where lower spatial aggregation is not useful. On the other side, the SW corner is more biased on the private view, where forecasts are needed for operational decisions, as vehicle routing & scheduling, and crowd shipping; symmetrically, high spatial aggregation is of no avail here.

The middle ground (mid-term forecasting) has only 3 papers (regional planning and emergency logistics): surprisingly, important tactical decisions, such as fleet & crew sizing, and location-routing are not mentioned in those

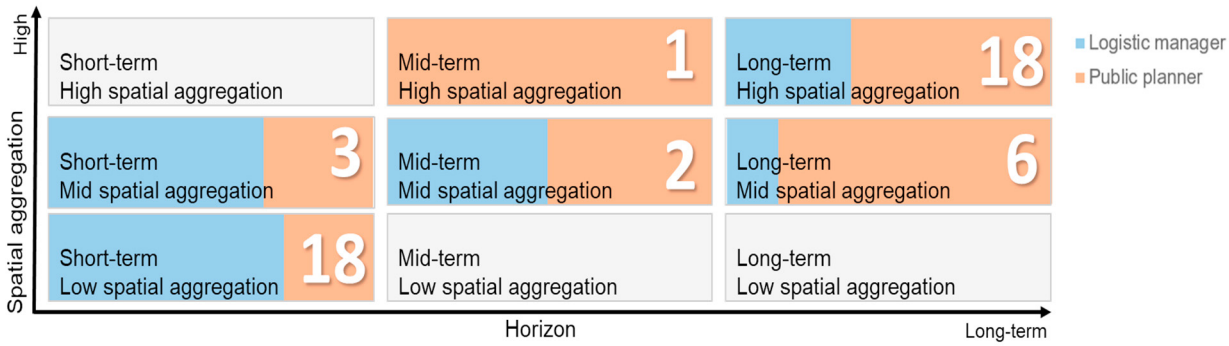

Fig. 4. Overview of horizon and spatial aggregation classifications according to stakeholder

papers.

4. Conclusion

This paper provides an overview of the characteristics of logistic demand forecasting models addressed in the literature based on a review of 48 papers. The results showed that research on forecasting has been developed by authors employing both public planners' and logistic managers' perspectives, although in different ways.

In general, authors with a logistics manager view are more interested in detailed forecasts to analyse demand fluctuations. This is observed by the predominance of short-term and low spatial aggregation articles. On the other hand, strategic decisions require a long-term forecast horizon, and this approach was also identified among the papers associated with the private sector. Distinctively, authors with a public planner point of view predominantly approach the topic of logistics demand forecasting in a more aggregated way - long term and high spatial aggregation. This difference may stem from the types of decisions that are considered by each of them.

It was observed that only works related to the public planner stakeholder used FG and FTG models, as they are very suitable to study certain city logistics problems, however no work of the private stakeholder used these models, although they could also be appropriate for this purpose. In view of this, future work may include terms derived from freight trip generation, such as freight attraction and freight production, for further analysis.

The literature on logistics demand forecasting is relatively scarce, and it was found that in the sample of selected papers there is a gap in mid-term forecasting. Direct data related to historical logistics demand is rarely available, making it necessary to model this problem from indirect sources and data, such as social, geographical, and economic indicators. Machine learning models were widely used in the sample of selected papers, especially in recent years. In addition, technological advances in the ability to obtain and process large volumes of data (disaggregated data) can bring in the future significant contributions to the decision-making process of private managers and public planners regarding the improvement of freight systems.

The proposed typology organizes and categorizes the articles, making it easier to identify relevant approaches, methods, and results for decision-makers in the public and private sectors. This allows for a clearer understanding of the problems that have already been explored and the gaps in logistics demand forecasting, providing a solid basis for the development of this topic in the literature and the application of these approaches in practice.

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Notes

Public repository of the 48 references: https://github.com/flaazevedo/logisticdemandreview

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